13

Polarons' Hall effect

As illustrated in Fig. 13.1, a Hall Effect results when a magnetic field H applied perpendicular to the current density J induces development of a Hall electric field F_H in the mutually perpendicular direction. These measurable parameters can be combined to yield the Hall coefficient $R_H \equiv F_H/JH$.

The transverse component of velocity $v_\perp \equiv \mu_H vH$ imparted to a carrier having velocity v by the magnetic field H defines the Hall mobility μ_H. The Hall angle $\theta_H \equiv v_\perp/v = \mu_H vH/v = \mu_H H$ describes this deflection. The Hall angle and Hall mobility are positive (indicating counterclockwise flow) for classical electrons. However, there can be no net dc steady-state carrier flow in a Hall measurement's transverse direction. Rather, space charge develops in the transverse direction so as to generate the Hall electric field. The presence of the Hall electric field rotates the net electric field thereby compensating the deflection of the current by the applied magnetic field: $F_H/(J/\sigma) = -\mu_H H$, where σ is the electrical conductivity. Inserting the definition of the Hall coefficient R_H yields an expression for the Hall mobility: $\mu_H = -\sigma R_H$. Thus the Hall mobility is obtained from measurements of σ and R_H.

Writing the electrical conductivity as the product of the carrier's density n, charge q and mobility μ, $\sigma = nq\mu$, enables the Hall coefficient to be expressed as $R_H = (1/nq)(-\mu_H/\mu)$. Here μ_H and μ are recognized as different measures of carriers' intrinsic (trap-free) motion. The Hall mobility μ_H measures the magnetic-field-induced deflection of carriers' flow. By contrast, the standard mobility μ measures carriers' movement in response to an electric field. The standard mobility has also been called the "conductivity mobility" and the "drift mobility". But the term "drift mobility" also refers to the trap-limited mobility measured in time-of-flight experiments. Since $-\mu_H/\mu$ is positive for quasi-classical carrier motion, the sign of its Hall coefficient $R_H = (1/nq)(-\mu_H/\mu)$ gives the sign of the carrier charge q. Measurement of R_H for quasi-classical carriers also provides an estimate of the carrier density n since then $\mu_H \approx -\mu$ and $R_H \approx 1/nq$.

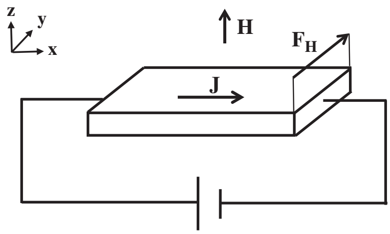

Fig. 13.1 The Hall electric field F_H develops across a material when the magnetic field H is applied perpendicular to the direction in which the current density J flows. For electrons' classical motion the Hall electric field is directed in the positive y-direction.

Polaron motion is usually qualitatively different from quasi-classical carrier motion. As a result, the magnitudes, temperature dependences and even the signs of polarons' Hall mobility are distinct from those of the quasi-classical carriers found in common semiconductors. Indeed, observing the distinctive features of polarons' Hall mobilities often facilitates identifying the carriers as polarons.

13.1 Coherent transport's Hall mobility

The Hall mobility can be calculated for coherent motion by solving the Boltzmann equation to first order in the strengths of both the electric and magnetic fields for the arrangement depicted in Fig. 13.1:

$$\mu_H = \frac{-q \sum\limits_{k_x,k_y} f_0(1-f_0)\left[\left(\dfrac{v_y^2}{m_{xx}} - \dfrac{v_x v_y}{m_{xy}}\right)\tau^2\right]}{\sum\limits_{k_x,k_y} f_0(1-f_0)\left(v_x^2\tau\right)}. \tag{13.1}$$

Here the distribution function for an equilibrated carrier of energy E is just the Fermi factor $f_0 = 1/\{\exp[(E - \mu_c)/\kappa T] + 1\}$, where μ_c denotes the carriers' chemical potential. The components of a carrier's velocity in the x- and y-directions, the partial derivatives of E with respect to the corresponding quasi-momentum $\hbar k_x$ or $\hbar k_y$, are denoted by v_x and v_y. Similarly, m_{xx} and m_{xy} are the components of the carrier's effective-mass tensor, the reciprocals of the second partial derivatives of E with respect to components of the carrier quasi-momentum. The carrier relaxation time is denoted by τ. All of these quantities, E, v_x, v_y, m_{xx}, m_{xy} and τ, are generally functions of k_x and k_y.

The Hall mobility, measuring the deflection of a current by a magnetic field, generally differs from the standard mobility, measuring the establishment of a

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current by an electric field. However, the magnitudes of these mobilities become comparable to one another in a semiconductor, $f_0\,(1-f_0) \rightarrow f_0 \approx \exp[-|E-\mu_c|/\kappa T] \ll 1$, with transport in a band that is much wider than κT. For example, $\mu_H = -\mu = -q\tau/m^*$ for charge transport involving states near a band extremum for which $|E| = m^*(v_x{}^2 + v_y{}^2 + v_z{}^2)/2$ when τ is approximated as a constant. Then the Hall coefficient becomes just $R_H = 1/nq$, where n is the density of mobile, untrapped, carriers.

A very different situation prevails if scattering-type transport involves an energy band whose width is considerably less than the thermal energy κT. In this limit f_0 becomes nearly independent of a carrier's wavevector. The mobility then reduces to

$$\mu = \frac{q}{\kappa T} \sum_{k_x} v_x^2 \tau. \tag{13.2}$$

Since each velocity component is proportional to corresponding bandwidth parameters the mobility is second order in them. Similarly, to second order in bandwidth parameters the Hall mobility of Eq. (13.1) becomes

$$\mu_H = \frac{-q \sum\limits_{k_x,k_y} \left[\left(\frac{v_y^2}{m_{xx}} - \frac{v_x v_y}{m_{xy}} \right) \tau^2 \right] \left(1 - \frac{E}{\kappa T} \right)}{\sum\limits_{k_x,k_y} \left(v_x^2 \tau \right)}, \tag{13.3}$$

where reciprocals of components of the effective mass tensor have been recognized as also being proportional to bandwidth parameters. In particular, the square-bracketed factor in the numerator is of third order in the bandwidth parameters while the denominator is of second order in them. The factor of E in the numerator raises it to fourth order in the bandwidth.

The temperature-independent contribution to the narrow-band limit of μ_H tends to vanish for the symmetric energy bands of highly symmetric structures. Thus, for example, a carrier's Hall mobility and the standard mobility are both temperature dependent in the narrow-band limit for square and face-centered cubic structures: $\mu_H = -\mu \propto 1/\kappa T$ when τ is assumed to be independent of the wavevector (Friedman, 1963). By contrast, the narrow-band limit for a carrier in a triangular lattice yields a Hall mobility that is temperature independent, larger than μ by the ratio of κT to the bandwidth, and can be anomalously signed (Friedman, 1963). The Hall constant then departs from its nominal value $1/nq$ in temperature dependence, magnitude and sign. Hall-effect sign anomalies occur when the contributions from states of negative effective mass dominate the summations of Eq. (13.3). For example, Hall-effect sign anomalies are found for coherent motion involving the narrow energy bands that are presumed to characterize anthracene's electronic transport (Friedman, 1964).

The Hall mobility of Eq. (13.1) applies to polarons whose inter-site motion is best described as being coherent but limited by occasional scattering events. As discussed in Section 10.1 large-polaron motion is generally coherent albeit with a narrow energy band. Such a large polaron's Hall mobility is described by Eq. (13.3).

This coherent transport picture has also been applied to a small polaron's motion in an ideal crystal at sufficiently low temperatures (Holstein, 1959b). With rising temperature the width of a small polaron's very narrow band decreases further and phonon scattering increases until coherent motion is destroyed. These two effects combine to preclude small-polarons' coherent motion in even an ideal crystal at all but low temperatures $\kappa T \ll \hbar\omega$ (Holstein and Friedman, 1968). Furthermore, as discussed in Section 11.1, coherent small-polaron motion is destroyed when even mild energetic disorder or the application of a modest electric field produces site-to-site energy differences that exceed the width of the extremely narrow small polaron band. For example, the small-polaron bandwidth for Holstein's molecular-crystal model is $\leq 6\hbar\omega\exp[-(E_b/\hbar\omega)\coth(\hbar\omega/2\kappa T)]$, where E_b is the small-polaron binding energy and $E_b \gg \hbar\omega$, cf. Section 4.4. Thus, in principle the small-polaron Hall mobility could be described with the narrow-band limit of Eq. (13.3) if the material is extremely well ordered, the temperature is very low and an applied electric field is very weak (Friedman, 1963).

In summary, the Hall mobility measures a different effect from that measured by the mobility which enters into the electrical conductivity. Nonetheless the magnitudes of these two mobilities usually become comparable to one another for coherent transport associated with an energy band whose width greatly exceeds the thermal energy κT. Consideration of the complementary limit, coherent transport associated with an energy band that is narrower than the thermal energy, reveals cases in which the two mobilities differ from one another in magnitude and temperature dependence. The sign of the Hall mobility can even be anomalous. Large polarons' motion is generally described as coherent. However, coherent small-polaron motion is only envisioned in extreme circumstances: low temperatures in nearly perfect crystals under the influence of weak applied fields. Otherwise small-polaron motion is described as being incoherent. The Hall effect for such hopping transport is the subject of the remainder of this chapter.

13.2 Microscopic treatment of electric and magnetic fields

Application of electric and magnetic fields alters a charge carrier's inter-site motion. In particular, the spatially varying electric and vector potentials associated with applied fields lift the translational degeneracy of a lattice's equivalent sites. Thus an applied electric field alters a carrier's potential energy as it moves between sites. As described below, a magnetic field's vector potential modifies a carrier's inter-site transfer integral.

For simplicity consider the tight-binding treatment of an excess electron in a crystal with atomic displacements being ignored. In the presence of a spatially constant magnetic field the local electronic function for an excess electron of energy $E_{\boldsymbol{g}}$ centered on lattice site $\boldsymbol{g}$ can be written as

$$\phi_{\boldsymbol{g}}(\boldsymbol{r}) = \exp\left[-i\frac{e}{2\hbar}(\boldsymbol{H}\times\boldsymbol{g})\cdot\boldsymbol{r}\right]\phi(\boldsymbol{r}-\boldsymbol{g}). \tag{13.4}$$

Here $q = -e$, where e is the electronic charge's magnitude and $\phi(\boldsymbol{r}-\boldsymbol{g})$ maintains the lattice periodicity by satisfying

$$\left\{\frac{1}{2m}\left[\hat{\boldsymbol{p}}+\frac{e}{2}\boldsymbol{H}\times(\boldsymbol{r}-\boldsymbol{g})\right]^2+V(\boldsymbol{r}-\boldsymbol{g})\right\}\phi(\boldsymbol{r}-\boldsymbol{g}) = E_{\boldsymbol{g}}\phi(\boldsymbol{r}-\boldsymbol{g}) \tag{13.5}$$

when the vector potential is written in its symmetrical gauge $\boldsymbol{A}(\boldsymbol{r}) = (\boldsymbol{H}\times\boldsymbol{r})/2$. Thus, with a spatially constant magnetic field a carrier's local electronic functions, the $\phi_{\boldsymbol{g}}(\boldsymbol{r})$'s, are distinguished from one another by their gauge-dependent phases.

The transfer integral linking local electronic functions at sites $\boldsymbol{g}$ and $\boldsymbol{g}'$ depends on the difference of their two magnetic-field-dependent phase factors: $\exp[-(ie/2\hbar)(\boldsymbol{g}-\boldsymbol{g}')\times\boldsymbol{r}\cdot\boldsymbol{H})]$, where vector algebra's cyclic rule has been applied. This transfer integral becomes proportional to $\exp[-(ie/4\hbar)(\boldsymbol{g}-\boldsymbol{g}')\times(\boldsymbol{g}+\boldsymbol{g}')\cdot\boldsymbol{H}]$ when its integrand peaks near $\boldsymbol{r} = (\boldsymbol{g}+\boldsymbol{g}')/2$. This factor simplifies to $\exp[-(ie/2\hbar)(\boldsymbol{g}\times\boldsymbol{g}')\cdot\boldsymbol{H}]$ after performing some vector algebra. Thus the energy associated with a carrier's transfer from site $\boldsymbol{g}$ to $\boldsymbol{g}'$ becomes $\exp(i\alpha_{\boldsymbol{g},\boldsymbol{g}'})t_{\boldsymbol{g},\boldsymbol{g}'}$, where $\alpha_{\boldsymbol{g},\boldsymbol{g}'} \equiv -(e/2\hbar)(\boldsymbol{g}\times\boldsymbol{g}')\cdot\boldsymbol{H}$ is termed the magnetic phase factor. When considering the Hall effect, which is first-order in the magnetic field strength, $t_{\boldsymbol{g},\boldsymbol{g}'}$ is usually taken as just the field-free electronic transfer energy. That is, corrections to $t_{\boldsymbol{g},\boldsymbol{g}'}$ that are first order in the magnetic field are usually ignorable. With sufficient symmetry these corrections even vanish (Friedman and Holstein, 1963).

Incorporating applied electric and magnetic fields $\boldsymbol{F}$ and $\boldsymbol{H}$ into the differential equations governing a carrier's set of site-occupation amplitudes, Eq. (11.21), the $a_{\boldsymbol{g}}$'s, yields:

$$i\hbar\frac{\partial a_{\boldsymbol{g}}}{\partial\tau} = \left(E_{\boldsymbol{g}}+e\boldsymbol{F}\cdot\boldsymbol{g}\right)a_{\boldsymbol{g}}+\sum_{\boldsymbol{g}'}\left(t_{\boldsymbol{g},\boldsymbol{g}'}e^{i\alpha_{\boldsymbol{g},\boldsymbol{g}'}}\right)a_{\boldsymbol{g}'}. \tag{13.6}$$

This equation incorporates a particular choice of the zero of the electric potential and of the gauge. However, physically observed quantities calculated from these equations are independent of these selections.

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Equation (13.6) provides the basis for describing the scattering-type transport addressed in Section 13.1. In particular, a carrier is represented by a wavepacket which evolves under the influence of electric and magnetic fields. Such a wavepacket is constructed from a superposition of Bloch states which are each expressed as a superposition of localized states. The evolution of the localized states' occupation amplitudes are governed by equations of motion, Eq. (13.6). This procedure regains the terms of the Boltzmann equation which describe carriers being driven by an electric field and deflected by a magnetic field via the Lorentz force (Friedman, 1963). The full Boltzmann equation is obtained upon adding its standard relaxation term. Thus the microscopic treatment of field-driven transport of Eq. (13.6) leads to Eq. (13.1), the solution of the Boltzmann equation, for scattering-type transport's Hall mobility.

13.3 Hall mobility for non-adiabatic polaron hopping

To address polaron motion in the presence of electric and magnetic fields the electronic equations of motion Eq. (13.6) must be augmented to incorporate atoms' movements. In particular, the Hamiltonian describing atoms' harmonic displacements is added to the electronic energy in Eq. (13.6), $E_{\boldsymbol{g}} \rightarrow E_{\boldsymbol{g}} + H_{\text{vib}}$:

$$i\hbar \frac{\partial a_{\boldsymbol{g}}}{\partial \tau} = \left(E_{\boldsymbol{g}} + H_{\text{vib}} + e\boldsymbol{F}\cdot\boldsymbol{g}\right) a_{\boldsymbol{g}} + \sum_{\boldsymbol{g}'} \left(t_{\boldsymbol{g},\boldsymbol{g}'} e^{i\alpha_{\boldsymbol{g},\boldsymbol{g}'}}\right) a_{\boldsymbol{g}'} . \tag{13.7}$$

The electron–phonon interaction manifests itself in the dependence of the local electronic energy $E_{\boldsymbol{g}}$ and the electronic transfer energies $t_{\boldsymbol{g},\boldsymbol{g}'}$ on atomic displacements. The microscopic equations of motion Eq. (13.7) provide the basis for discussing the Hall effect for polaron hopping.

The Hall effect for hopping polarons manifests itself through jump rates' dependences on the strengths of the applied electric and magnetic fields. These jump rates are calculated from the equations of motion Eq. (13.7). Jump rates have been computed in two limits. In the adiabatic limit the electronic transfer energies are assumed large enough for an electronic carrier to adjust to atoms' movements. A carrier will then always negotiate a hop in response to appropriate atomic motions. In the non-adiabatic limit the electronic transfer energies are assumed to be so small that an electronic carrier can only rarely negotiate a hop in response to appropriate atomic motions. Electronic transfer energies are often large enough (typically > $\hbar\omega$ and κT) to justify employing the adiabatic limit. However, calculations are easiest in the non-adiabatic limit. Then Eq. (13.7) is solved by treating electronic transfer energies as small perturbations which enable the excess electron to jump between sites.

The simplest geometry for which to address the Hall mobility for polaron hopping is the triangular lattice depicted in panel *a* of Fig. 13.2. This geometry was first used to address the Hall effect for non-adiabatic polaron hopping (Friedman and Holstein, 1963). To lowest order in the transfer energies the non-adiabatic jump rate is proportional to the square of the absolute value of transfer energy connecting the hop's two sites, $|t_{\mathbf{g},\mathbf{g}'} \exp(i\alpha_{\mathbf{g},\mathbf{g}'})|^2$. Since the magnetic phase factors do not contribute to the absolute value, this term does not contribute to the jump rate's magnetic-field dependence. However, processes that are of higher order in the electronic transfer energies contribute to the magnetic-field dependence of the jump rate. In particular, the lowest-order magnetic-field dependent contributions to the jump rate in the triangular lattice are third order in the electronic transfer energies. These processes, illustrated in panel *a* of Fig. 13.3, involve the interference between the amplitude for a carrier's direct hop between adjacent sites ($\mathbf{g} \rightarrow \mathbf{g}'$) and the amplitude for the carrier's passing through an intermediate site ($\mathbf{g} \rightarrow \mathbf{g}'' \rightarrow \mathbf{g}'$). The magnetic-field dependent portion of the jump rate is proportional to the sum of the magnetic phase factors for the three involved jumps: $\alpha_{\mathbf{g},\mathbf{g}''} + \alpha_{\mathbf{g}'',\mathbf{g}'} - \alpha_{\mathbf{g},\mathbf{g}'}$. Utilizing the definition of the magnetic phase factors from above Eq. (13.6) and elementary rules of vector algebra, the sum of the magnetic phase factors is found to equal $e\mathbf{H}\cdot\mathbf{A}_\Delta/\hbar$, where $\mathbf{A}_\Delta \equiv (\mathbf{g}'' - \mathbf{g}) \times (\mathbf{g}' - \mathbf{g})/2$ is the area of the triangle subtended by the three involved sites. Thus, the magnetic-field-dependent portion of the non-adiabatic jump rate is proportional to the magnetic flux passing through the triangle defined by sites $\mathbf{g}$, $\mathbf{g}'$ and $\mathbf{g}''$.

Studies of the Hall effect for polaron hopping focused on the high-temperature regime $\kappa T > \hbar\omega$ in which atomic motion becomes classical. Atoms' movements

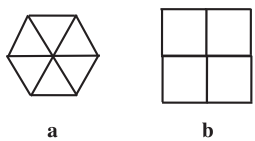

Fig. 13.2 Portions of a triangular lattice and a square lattice are illustrated in panels *a* and *b*, respectively. Hops occur between sites located at these structures' vertices.

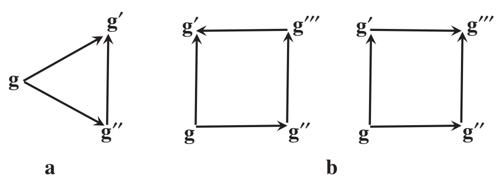

Fig. 13.3 The three-site and four-site interference processes which generate the predominant magnetic-field dependences for non-adiabatic nearest-neighbor polaron hopping in triangular and square lattices are illustrated in panels *a* and *b*, respectively.

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are then treated as explicit functions of time. In this regime contributions to the polaron jump rate are thermally activated. The activation energy for the leading field-free contribution to the jump rate is the minimum energy required to displace atoms from the polaron's equilibrium configuration to bring the electronic energies of a hop's initial and final sites into coincidence with one another E_2 (cf. Section 11.5). This two-site activation energy is that for the standard mobility $E_2 = E_A$. The activation energy for the leading magnetic-field-dependent contribution to the jump rate is the minimum energy needed to bring the electronic energies of its three sites into coincidence E_3. Since the constraint for a three-site coincidence exceeds that for a two-site coincidence $E_3 > E_2$. As described in the second paragraph of this chapter, the Hall mobility is the magnetic-field-induced transverse component of a carrier's velocity divided by the product of its longitudinal component and the magnetic field. Thus, the Hall mobility's activation energy is $E_H \equiv E_3 - E_2$. For the short-range electron–phonon interaction envisioned in Holstein's molecular-crystal model $E_n = (F^2/2k)[(n - 1)/n]$ and $E_H \equiv (E_3 - E_2) = E_2/3$.

The lowest-order magnetic-field-independent (second-order) and magnetic-field-dependent (third-order) contributions to the non-adiabatic jump rate for a triangular lattice are (Friedman and Holstein, 1963):

$$R_2 = \left[\frac{t^2}{\hbar}\left(\frac{\pi}{4\kappa T E_2}\right)^{1/2}\right] e^{-E_2/\kappa T} \tag{13.8}$$

and

$$R_3 = \left(\frac{e\boldsymbol{H}\cdot\boldsymbol{A}_\Delta}{\hbar}\right)\left[\frac{2}{\sqrt{3}}\frac{t^3}{\hbar}\left(\frac{\pi}{4\kappa T E_2}\right)\right] e^{-E_3/\kappa T}, \tag{13.9}$$

where the field-free geometrically equivalent nearest-neighbor transfer energies are written as $t_{\boldsymbol{g},\boldsymbol{g}'} \equiv -t$. As described in Section 11.5, for non-adiabatic hopping (1) the duration of a coincidence τ_c is much less than a vibrational period $\omega\tau_c \ll 1$ and (2) the time needed for a carrier's transfer $\tau_t \equiv \hbar/|t|$ is much longer than the time window provided by a coincidence event $\tau_t \gg \tau_c$. Expressed in terms of these times $R_2 \approx \nu(\tau_c/\tau_t)^2\exp(-E_2/\kappa T)$ and $R_3 \approx \nu(e\boldsymbol{H}\cdot\boldsymbol{A}_\Delta/\hbar)(\tau_c/\tau_t)^3(\omega\tau_c)\exp(-E_2/3\kappa T)$, where $\nu \equiv \omega/2\pi$. The factor $\omega\tau_c$ in R_3 arises from the requirement that atomic vibrations bring the electronic energies of all three sites into coincidence within the time interval τ_c.

The second-order jump rate is simply related to the in-plane mobility for the triangular lattice:

$$\mu_{\rm t} = -\frac{3}{2}\frac{ea^2}{\kappa T}R_2 = -\left(\frac{ea^2}{\hbar}\right)\left[\frac{3t^2}{2\kappa T}\left(\frac{\pi}{4\kappa TE_2}\right)^{1/2}\right]e^{-E_2/\kappa T}. \tag{13.10}$$

As described in the second paragraph of Section 13.1, the Hall mobility measures the deflection by the magnetic field of carriers drifting in an electric field. As a result, the Hall mobility of a carrier hopping in a triangular lattice depends on the ratio R_3/R_2 (Friedman and Holstein, 1963):

$$\mu_{H,\rm t} = \frac{2}{\sqrt{3}}\frac{R_3}{H_\perp R_2} = \left(\frac{ea^2}{\hbar}\right)\left[\frac{t}{\sqrt{3}}\left(\frac{\pi}{4\kappa TE_2}\right)^{1/2}\right]e^{-(E_3-E_2)/\kappa T}, \tag{13.11}$$

with $E_3 - E_2 = E_2/3$ for the molecular-crystal model. In obtaining Eq. (13.11) the magnetic flux through the triangle $\boldsymbol{H}\cdot\boldsymbol{A}_\Delta$ was written $H_\perp A_\Delta$, where $H_\perp$ is the component of the magnetic field perpendicular to the current flow and the area of the triangular lattice's equilateral triangle is expressed in terms of the inter-site separation a as $A_\Delta = \sqrt{3}a^2/4$. The Hall coefficient is then

$$R_{H,\rm t} = -\frac{\mu_{H,\rm t}}{\sigma} = \frac{1}{ne}\left(\frac{\mu_{H,\rm t}}{\mu_{\rm t}}\right) = \frac{-1}{ne}\left[\left(\frac{2}{3\sqrt{3}}\right)\frac{\kappa T}{t}\right]e^{(2E_2-E_3)/\kappa T}, \tag{13.12}$$

where $2E_2 - E_3 = 2E_2/3$ for Holstein's molecular-crystal model (Friedman and Holstein, 1963). The Hall coefficient for non-adiabatic polaron hopping in the triangular lattice is seen to depart significantly in magnitude and temperature dependence from its free-carrier value, $-1/ne$.

The magnitudes of the non-adiabatic mobility and Hall mobility for the triangular lattice are both plotted against reciprocal temperature in Fig. 13.4. These plots indicate that the Hall mobility is generally larger than the standard mobility. The temperature dependence of the Hall mobility is also much weaker than that of the standard mobility.

The Hall mobility has also been calculated for non-adiabatic semiclassical polaron hopping between sites of the square lattice depicted in panel *b* of Fig. 13.2 (Emin, 1971c). The Hall effect for non-adiabatic hopping results from interferences between the transition amplitudes associated with different hopping routes that connect the same initial and final sites. For nearest-neighbor hopping on a square lattice these interference processes require at least four transfers. The corresponding magnetic-field-dependent jump rates are therefore of fourth order in the electronic-transfer energies. Both types of interference process are also proportional to the sums of the process' magnetic phase factors: $\alpha_{g,g''} + \alpha_{g'',g'''} + \alpha_{g''',g'} - \alpha_{g,g'}$ and $\alpha_{g,g''} + \alpha_{g'',g'''} - \alpha_{g,g'} - \alpha_{g',g'''}$, respectively. Both of these sums of magnetic phase factors equal $e\boldsymbol{H}\cdot\boldsymbol{A}_\square/\hbar$,

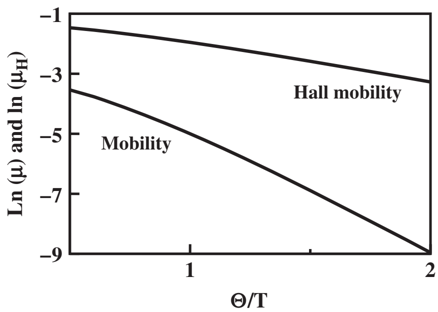

Fig. 13.4 The magnitudes of the non-adiabatic mobility and non-adiabatic Hall mobility in units of $ea^2/\hbar$ for a triangular lattice are plotted versus reciprocal temperature in units of the phonon temperature $\Theta \equiv \hbar\omega/\kappa$. These plots' parameters are $E_2 = 5\hbar\omega$ and $t = \hbar\omega/2$.

where $\boldsymbol{A}_\square \equiv (\boldsymbol{g}''' - \boldsymbol{g}) \times (\boldsymbol{g}' - \boldsymbol{g}'')/2$ is the area of the square enclosed by the four involved sites, a^2 for inter-site separation a. Thus, the magnetic-field-dependent portions of the non-adiabatic jump rates for a square lattice are proportional to the magnetic flux passing through one of the lattice's elementary units.

As in the case of the triangular lattice, the magnetic-field-dependent contribution to the non-adiabatic semiclassical jump rate for a square lattice is associated with the formation of a triple coincidence. In these instances the electronic energies of initial and final sites of the interference process share a transitory equality with one of the two other sites. The fourth-order magnetic-field-dependent contribution to jump rates in the square lattice is similar to that for the triangular lattice, Eq. (13.9):

$$R_4 = \left(\frac{eH_\perp a^2}{\hbar}\right)\left[\frac{2}{\sqrt{3}}\frac{t^3}{\hbar}\left(\frac{\pi}{4\kappa T E_2}\right) I_4\right] e^{-E_3/\kappa T}, \tag{13.13}$$

where $H_\perp$ is the component of the magnetic field perpendicular to the plane of the square lattice and the effect of the non-coincident site is contained in the factor I_4. The Hall mobility for non-adiabatic nearest-neighbor hopping in a square lattice also resembles the analogous expression for a triangular lattice, Eq. (13.11):

$$\mu_{H,\mathrm{s}} = \frac{4R_4}{H_\perp R_2} = \left(\frac{ea^2}{\hbar}\right)\left[\frac{t}{\sqrt{3}}\left(\frac{\pi}{4\kappa T E_2}\right)^{1/2}(8I_4)\right] e^{-(E_3 - E_2)/\kappa T}. \tag{13.14}$$

The Hall mobility of a square lattice is distinguished from that of a triangular lattice by the presence of the non-coincident site. As illustrated in Fig. 13.5, virtual occupations of non-coincident sites provide bridges which enable transfer among

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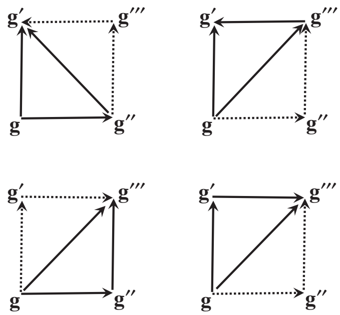

Fig. 13.5 The magnetic-field-dependent contribution to the semiclassical jump rate between sites requires forming a triple coincidence involving the electronic states at these two sites plus that associated with one of the intermediate sites. Virtual transfer through non-coincident sites, depicted with dotted arrows, enable effective transfers, represented by solid arrows, among the three coincident sites. The four pertinent situations relevant to four-site hopping in a square lattice are illustrated above.

trios of coincident sites. The sign and magnitude of a virtual transfer through a non-coincident site depend on the difference between its electronic energy and the electronic energy of the three coincident sites. This difference of electronic energies depends on atomic displacements. In particular, the sign of the virtual transfer changes when alterations of atomic displacements cause the electronic energy of the non-coincident site to overtake the electronic energy of the three coincident sites. This sign change occurs when the four sites experience a quadruple coincidence.

The minimum atomic-displacement energy needed to establish a quadruple-coincidence is denoted E_4. The energy characterizing a deformation-related change of the sign of the virtual transfer involved in the semiclassical Hall effect is $E_4 - E_3$. This energy difference is generally a small fraction of E_2. For example, $E_4 - E_3 = E_2/6$ for Holstein's molecular-crystal model. The mobility activation energy E_2 is frequently estimated from experiment. In many instances $E_4 - E_3$ is less than the thermal energy κT where mobility measurements are performed.

The introduction of the factor I_4 in Eq. (13.13) and Eq. (13.14) [as compared with Eq. (13.9) and Eq. (13.11)] results from the elementary structural unit of the square lattice having one more site than that of the triangular lattice. In particular, the virtual transfer energy averaged over thermally induced atomic displacements is essentially tI_4 (Emin, 1971c). For classical atomic motion $I_4 \to t/E_3$ when $E_4 - E_3 \gg \kappa T$. Then $\mu_{H,\mathrm{s}}$ only differs from $\mu_{H,\mathrm{t}}$ by a temperature-independent factor. By contrast, $I_4 \to (t/4\kappa T)\exp[-(E_4 - E_3)/\kappa T]$ when $E_4 - E_3 < \kappa T$. In these instances $\mu_{H,\mathrm{s}}$ manifests a different temperature dependence than $\mu_{H,\mathrm{t}}$. This feature is evident in Fig. 13.6 where $\mu_{H,\mathrm{t}}$ and $\mu_{H,\mathrm{s}}$ are both plotted against reciprocal temperature.

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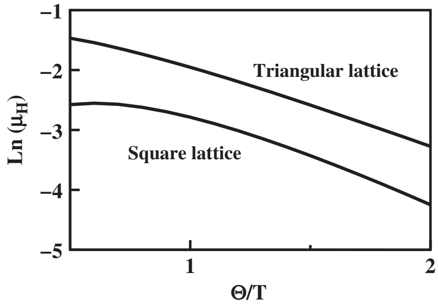

Fig. 13.6 The non-adiabatic Hall mobilities in units of $ea^2/\hbar$ for triangular and square lattices are plotted versus reciprocal temperature in units of the phonon temperature $\Theta \equiv \hbar\omega/\kappa$. These plots' parameters are $E_2 = 5\hbar\omega$ and $t = \hbar\omega/2$.

The Hall coefficient for nearest-neighbor hopping on a square lattice becomes:

$$R_{H,\mathrm{s}} = -\frac{\mu_H}{\sigma} = \frac{1}{ne}\frac{\mu_{H,\mathrm{s}}}{\mu_\mathrm{s}} = \frac{-1}{ne}\left[\frac{\kappa T}{t}\frac{8I_4}{\sqrt{3}}\right]e^{(2E_2-E_3)/\kappa T} \tag{13.15}$$

upon recalling the Hall mobility Eq. (13.14) and noting that a square lattice's mobility is

$$\mu_\mathrm{s} = -\frac{ea^2}{\kappa T}R_2 = -\left(\frac{ea^2}{\hbar}\right)\left[\frac{t^2}{\kappa T}\left(\frac{\pi}{4\kappa T E_2}\right)^{1/2}\right]e^{-E_2/\kappa T}. \tag{13.16}$$

When $\kappa T > E_4 - E_3$ the Hall coefficient rises exponentially with increasing reciprocal temperature, $R_{H,\mathrm{s}} \sim (-1/ne)\exp[(2E_2 - E_4)/\kappa T]$ with $2E_2 > E_4$. Similar behavior persists even if the electron–phonon coupling is so strong that $E_4 - E_3 \gg \kappa T$: $R_{H,\mathrm{s}} \sim [(-1/ne)(5\kappa T/E_3)\exp[(2E_2 - E_3)/\kappa T]$ with $2E_2 > E_3$. Thus the mobility ratio produces a temperature dependence which mimics that from a semiconductor's thermally activated carrier density.

All told, the non-adiabatic limit of semiclassical polaron hopping produces a distinctive Hall effect. The Hall effect arises as atoms' vibrations cause the electronic energies of a hop's initial and final sites to come into momentary coincidence with the electronic energy of another site. The magnitude of the resulting Hall mobility is generally larger than that of the conventional mobility. In addition, the Hall mobility's temperature dependence is markedly weaker than the thermally activated behavior of the standard hopping mobility. The large disparity between the temperature dependences of the Hall mobility and the conventional mobility leads to a Hall coefficient whose temperature dependence often resembles that arising from a thermally activated carrier density. These distinguishing features can be utilized to identify polaron hopping.

13.4 Hall mobility for adiabatic polaron hopping

Hopping is non-adiabatic when the governing electronic transfer energy is so small that electronic carriers usually cannot transfer fast enough to negotiate a hop in response to suitable atomic motion. Such electronic transfer energies are typically less than the characteristic phonon energy, $t < \hbar\omega$. By contrast, adiabatic hopping occurs when electronic transfer energies are large enough to enable carriers to follow atomic movements which are themselves modified by the electronic carriers. The Hall effect has been calculated for small-polaron hopping among sites of Holstein's molecular-crystal model arranged in a triangular lattice (Emin and Holstein, 1969).

An applied magnetic field generally alters a carrier's wavefunction. As described in Section 13.2 this change causes the carrier's electronic transfer energies to garner magnetic-field-dependent phase factors. These magnetic phase factors affect the adiabatic Hamiltonian governing atomic motion. In particular, these magnetic phase factors generate those terms in the adiabatic Hamiltonian of Eq. (4.5) which are proportional to the momenta $\boldsymbol{P}$ that are conjugate to atomic displacement parameters associated with the reduced mass M. As shown in Eq. (4.5) these terms have the form $e(\boldsymbol{A}\cdot\boldsymbol{P}+\boldsymbol{P}\cdot\boldsymbol{A})/2M$, where Eq. (4.6) defines the fictitious vector potential $\boldsymbol{A}$. This fictitious vector potential is proportional to the actual applied magnetic field $\boldsymbol{H}$.

Hopping transport envisions jumps of a carrier between discrete sites. Therefore a carrier's location is indicated by the centroid of the occupied site $\boldsymbol{g}$ rather than by the continuous variable $\boldsymbol{r}$. Concomitantly, the carrier's wavefunction is designated by the set of occupation amplitudes $\{c_{\boldsymbol{g}}\}$ rather than by the continuous function $\phi(\boldsymbol{r})$ used in Eq. (4.6). Furthermore, the strain $\boldsymbol{\Delta}(\boldsymbol{u})$ at position $\boldsymbol{u}$ which characterizes the deformable continuum described in Chapter 4 is replaced by a set of local deformation parameters $\{x_{\boldsymbol{g}}\}$. For Holstein's molecular-crystal model the energy of a carrier at site $\boldsymbol{g}$ is presumed to depend linearly on the deformation parameter which is assigned to the same site: $E_{\boldsymbol{g}} \equiv -Fx_{\boldsymbol{g}}$. The scalar F then denotes the magnitude of this short-range electron–phonon interaction. Using Eq. (13.6) the electronic amplitudes corresponding to the electronic eigenvalue $W(\ldots\, x_{\boldsymbol{g}} \,\ldots)$ are found to satisfy

$$Wc_{\boldsymbol{g}} = \left(-Fx_{\boldsymbol{g}} + e\boldsymbol{F}\boldsymbol{\cdot}\boldsymbol{g}\right)c_{\boldsymbol{g}} + \sum_{\boldsymbol{g}'}\left(t_{\boldsymbol{g},\boldsymbol{g}'}e^{i\alpha_{\boldsymbol{g},\boldsymbol{g}'}}\right)c_{\boldsymbol{g}'}. \tag{13.17}$$

A carrier executes a thermally activated semiclassical hop from site $\boldsymbol{g}$ to site $\boldsymbol{g}'$ when atomic deformation parameters change in time so that $|c_{\boldsymbol{g}}|^2$ falls from near unity to near

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zero as $|c_{g'}|^2$ rises from near zero to near unity. These changes of sites' occupation probabilities primarily occur when the electronic energies of the two sites are within $|t_{g,g'}| \equiv t$ of being coincident with one another. The transfer energy t is generally smaller than the deformation energy associated with establishing a coincidence $F^2/4k$ since small-polaron formation requires that $F^2/2k \gg t$. Alternatively stated, the conformational change characterizing a coincidence's establishment $F/4k$ exceeds that characterizing an electronic carrier's transfer t/F. In addition, the deformation energy associated with forming a coincidence is usually much larger than the thermal energy: $F^2/4k \gg \kappa T$. In other words, a coincidence's characteristic conformal change is much larger than atomic vibrations' thermal amplitudes: $F/4k \gg (\kappa T/k)^{1/2}$, where k is the deformation parameter's stiffness constant. Thus, atomic vibrations only sporadically enable a small polaron to execute an adiabatic hop.

The Hall effect for nearest-neighbor hopping in a triangular lattice can be addressed by considering hopping among three mutually adjacent sites. A Hall effect results when the magnetic field alters atomic motion so as to change the final site of a hop. Then, for example, a carrier which would have hopped from site 1 to site 2 in the absence of the magnetic field jumps to site 3 when the magnetic field is applied.

The relative values of the three sites' atomic displacement parameters determine which site the carrier occupies. An orthogonal transformation of the molecular-crystal-model's deformation parameters for the three sites, x_1, x_2 and x_3, yields two relative displacement coordinates, x and y:

$$x \equiv \frac{x_2 + x_3 - 2x_1}{\sqrt{6}} \tag{13.18}$$

and

$$y \equiv \frac{x_3 - x_2}{\sqrt{2}}, \tag{13.19}$$

with $z \equiv (x_1 + x_2 + x_3)/\surd 3$. Since c_1, c_2 and c_3 only depend on x and y, semiclassical hops are described by these variables' evolution with time τ. The classical trajectories, $x(\tau)$ and $y(\tau)$, are governed by the adiabatic Hamiltonian. The adiabatic Hamiltonian for the three pertinent sites contains a vibratory contribution $k(x^2 + y^2 + z^2)/2$ and an electronic contribution $W = \varepsilon(x, y) - Fz/\surd 3$.

Three electronic eigenstates are found by solving Eq. (13.17) for three mutually adjacent sites. Upon taking the nearest-neighbor transfer energies to be negative, $t_{g,g'} = -t$ with $t > 0$, the lowest energy electronic eigenstate is found to be nodeless and non-degenerate. The corresponding energy $\varepsilon(x,y)$ then comprises a triangular pyramid whose edges become rounded for finite t. This rounding of the pyramid's apex

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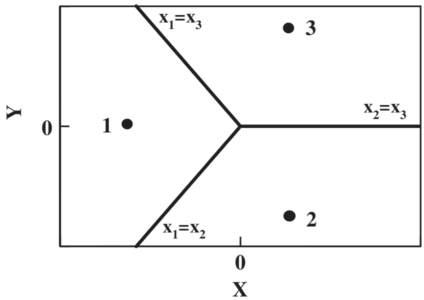

Fig. 13.7 The space defined by the configuration coordinates x and y consists of regions labeled as 1, 2, and 3. Each region's label indicates the site occupied by the carrier for configuration coordinates located in that region. The boundaries between the three regions are defined by coincidence conditions between the corresponding sites' deformation parameters. The minimum of the adiabatic potential within each region is indicated with a large dot.

reduces its energy from $\varepsilon(0,0) = 0$ to $\varepsilon(0,0) = -2t$. As illustrated in Fig. 13.7 the three edges of the triangular pyramid coincide with the coincidence lines $x_1 = x_2\left(y = x\sqrt{3} \text{ for } x \le 0\right)$, $x_1 = x_3\left(y = -x\sqrt{3} \text{ for } x \le 0\right)$, and $x_2 = x_3$ ($y = 0$ for $x \ge 0$). These three coincidence lines divide the xy configuration space into three equivalent regions within which the occupation probabilities, the $|c_g|^2$, for each of the three corresponding sites approach unity. Each region is therefore labeled by the corresponding occupied site.

In the limit $t \to 0$ the rounding of the pyramid's edges disappears. Then the pyramidal electronic energy contour assumes a simple form:

$$\varepsilon(x,y) = \frac{F}{\sqrt{6}}(2x), \tag{13.20}$$

$$\varepsilon(x,y) = -\frac{F}{\sqrt{6}}\left(x - \sqrt{3}y\right) \tag{13.21}$$

and

$$\varepsilon(x,y) = -\frac{F}{\sqrt{6}}\left(x + \sqrt{3}y\right) \tag{13.22}$$

in regions 1, 2 and 3, respectively. Combining $\varepsilon(x, y)$ with the vibrational potential $k(x^2 + y^2)/2$ yields a potential well with minima of $-F^2/3k$ located in each of the three regions. These minima are at $\left(-2F/k\sqrt{6}, 0\right)$, $\left(F/k\sqrt{6}, -F/k\sqrt{2}\right)$ and $\left(F/k\sqrt{6}, F/k\sqrt{2}\right)$ in regions, 1, 2, and 3, respectively. A semiclassical adiabatic hop between sites

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is represented by a trajectory which passes between the corresponding potential wells.

A reduced adiabatic Hamiltonian governs the atomic motions which dictate the excess electron's movement among three mutually adjacent sites of a triangular molecular-crystal lattice:

$$H_3 = \frac{P_x^2 + P_y^2}{2M} + \frac{k}{2}\left(x^2 + y^2\right) + \varepsilon(x, y) + \frac{e}{2M}(\boldsymbol{A} \cdot \boldsymbol{P} + \boldsymbol{P} \cdot \boldsymbol{A}). \tag{13.23}$$

The first three contributions are (1) the kinetic-energy associated with movement within the xy configuration coordinate plane, (2) the potential energy of these configuration coordinates' vibrations and (3) the electron's energy as a function of these configuration coordinates. The remaining term is central to the Hall effect. It couples the momenta conjugate to the x and y configuration coordinates with the fictitious vector potential. The fictitious vector potential results from the magnetic-field dependence of the electronic carrier's inter-site transfer in response to changes of the configuration coordinates. The magnetic field generated from the fictitious vector potential is (1) proportional to the applied magnetic field, (2) directed perpendicular to the xy plane, and (3) strongly peaked about the origin, the triple-coincidence point. The decrease of the fictitious magnetic field's strength with distance from the origin is (1) generally characterized by the configuration coordinate distance $\sim t/F$ and (2) weakest along the three double coincidence lines. The total flux of the fictitious magnetic field through the xy configuration coordinate plane equals that of the applied magnetic field through the real-space triangle whose vertices are three mutual nearest-neighbor sites (Emin and Holstein, 1969).

Semiclassical adiabatic hops between sites are depicted by trajectories which pass between potential wells in the xy configuration space. Thus an adiabatic hop from site 1 to site 3 corresponds to a trajectory which passes from the potential well of region 1 to that of region 3. These excursions are governed by the Hamiltonian of Eq. (13.23). Vibrational dispersion is not explicitly present in this reduced three-site Hamiltonian. Nonetheless its implicit presence enables the fictitious particle to exchange energy with the remainder of the lattice before and after a hop. As discussed in Section 11.6, this energy transfer is presumed sufficient for each inter-well passage to be regarded as an independent event. Thus the fictitious particle whose trajectory describes a thermally assisted hop usually garners energy before leaving its initial region and deposits energy after entering its final region.

The adiabatic jump rate associated with a pair of sites is the probability per unit time of a trajectory passing between the corresponding regions. To count the relevant trajectories it is useful to construct a reference line deep within the region

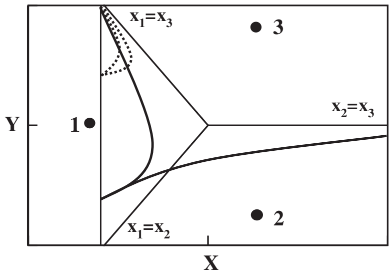

Fig. 13.8 At low energies the trajectories from a position on the reference line which can reach region 3 are bracketed by two trajectories (dotted lines) which are tangent to the $x_1 = x_3$ coincidence line. At high energies the trajectories from a position on the reference line which can reach region 3 are bracketed by one solid-line trajectory which is tangent to the $x_1 = x_3$ coincidence line and another solid-line trajectory which is tangent to the $x_2 = x_3$ coincidence line.

corresponding to the trajectories' initial site. The reference line for trajectories leaving region 1 is the line $x = -d$, where $2F/k\sqrt{6} > d \gg t/F$. A trajectory leaving region 1 can be characterized by its energy E and by the transverse component of velocity v_y it possesses as it crosses the reference line. Then, as illustrated in Fig. 13.8, trajectories of energy E crossing the reference line of region 1 at $y = y_0$ will reach region 3 for values of v_y between $v_y^{\max}$ and $v_y^{\min}$. If E is less than that required for trajectories to reach the peak of the adiabatic potential at $x = y = 0$ then both of these limiting trajectories will be tangential to the $x_1 = x_3$ coincidence line. However, if E is sufficient for trajectories to reach the peak of the adiabatic potential then one of the limiting trajectories for passing from region 1 to region 3 will be tangential to the $x_1 = x_3$ coincidence line while the other is tangential to the $x_2 = x_3$ coincidence line.

The application of a magnetic field shifts the limiting trajectories from each point on the reference line (Emin and Holstein, 1969). In particular, an arbitrarily weak magnetic field just alters the y-components of the fictitious-particle's velocities for the two limiting trajectories which cross the reference line at y_0 by $\Delta v_y^{\max}$ and $\Delta v_y^{\min}$. The net change of the probability per unit time and per unit energy passing from region 1 to region 3 induced by the magnetic field is then

$$W_{13}^{H}(E) = \frac{e^{-E/\kappa T}}{MZ} \int_{y_1}^{y_2} dy_0 \left(\Delta v_y^{\max} - \Delta v_y^{\min}\right). \tag{13.24}$$

Here $y_2(E)$ and $y_1(E)$ are the limits of y_0 between which trajectories of energy E reach region 3. The partition function for a mass M vibrating with frequency $\nu \equiv (k/M)^{1/2}/2\pi$ within the quasi-harmonic potential well of depth $-F^2/3k = -E_3$ located in region 1 is

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$$Z \approx \left(\frac{\kappa T}{M v}\right)^2 \exp(E_3/\kappa T). \tag{13.25}$$

Here E_3 is recognized as the three-site coincidence energy for Holstein's molecular-crystal model defined in the paragraph before Eq. (13.8).

A theorem enables the magnetic-field-dependent contribution to the jump probability from trajectories of energy E, Eq. (13.24), to be expressed in terms of the flux of the fictitious magnetic field enclosed by limiting trajectories of energy E (Emin and Holstein, 1969). In particular,

$$\int_a^b dy_0 \Delta v_y = -\oint_C d\boldsymbol{l}\cdot\Delta\boldsymbol{v} = \frac{-e}{M}\Phi_C(E), \tag{13.26}$$

where $\Phi_C(E)$ is the flux of the fictitious magnetic field enclosed by the contour of integration C. The integration contour follows (1) the limiting trajectory from $y_0 = a$ on the reference line to asymptotically approach a coincidence line, (2) the limiting asymptotic trajectory from this coincidence line to the reference line at $y_0 = b$ and (3) the reference line from $y_0 = b$ to $y_0 = a$.

This theorem enables Eq. (13.24) to be expressed in terms of $\Phi(E)$, the flux enclosed by limiting trajectories (Emin and Holstein, 1969):

$$W_{13}^H(E) = e\left(\frac{v}{\kappa T}\right)^2 e^{-(E_3+E)/\kappa T}\Phi(E). \tag{13.27}$$

When $E < -2t$, the energy of the triplet-coincidence peak, trajectories lack sufficient energy to reach the $x_2 = x_3$ coincidence line from the region 1 reference line. Then $\Phi(E) = 0$ as all limiting trajectories are tangent to the $x_1 = x_3$ coincidence line. By contrast, for $E \geq -2t$ some trajectories can reach the $x_1 = x_3$ coincidence line. As illustrated in Fig. 13.9 $\Phi(E)$ is then the flux enclosed between the three equivalent limiting trajectories which are each tangent to two coincidence lines. The enclosed flux increases with energy as the limiting trajectories move further from the triple-coincidence point to encompass an increasing area of the xy plane.

Integrating $W_{13}^H(E)$ over energy yields the magnetic-field-dependent contribution to the adiabatic jump rate in a triangular lattice, analogous to R_3 of Eq. (13.9) for non-adiabatic hopping:

$$R_3^{\mathrm{A}} = (e\boldsymbol{H}\cdot\boldsymbol{A}_\Delta)\left(\frac{v^2}{\kappa T}\right)e^{-(E_3-2t)/\kappa T}F(T), \tag{13.28}$$

where

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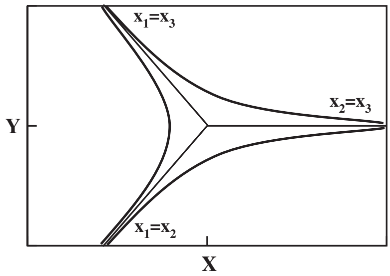

Fig. 13.9 Three equivalent limiting trajectories that are each tangent to two coincidence lines define the border which encloses the magnetic flux $\Phi(E)$. These trajectories move further from the coincidence lines and from the triple coincidence point as their energy E increases. The effective magnetic field's strength is greatest along the coincidence lines and has its absolute maximum at the triple coincidence point. Therefore the enclosed magnetic flux rises as these limiting trajectories' energy is increased.

$$F(T) \equiv \int_{-2t}^{\infty} \frac{dE}{\kappa T} e^{-(E+2t)/\kappa T} \frac{\Phi(E)}{\boldsymbol{H} \cdot \boldsymbol{A}_{\Delta}}. \tag{13.29}$$

This function increases monotonically from zero to unity as $\kappa T/t$ is raised from well below 1 to well above 1 (Emin and Holstein, 1969). The component of the adiabatic jump rate which is independent of the magnetic field is just

$$R_2^{\mathrm{A}} = \nu e^{-(E_2-t)/\kappa T}. \tag{13.30}$$

These results are utilized to obtain the adiabatic Hall and standard mobilities for semiclassical nearest-neighbor polaron hopping in the triangular lattice:

$$\mu_{H,\mathrm{t}}^{\mathrm{A}} = \frac{2}{\sqrt{3}} \frac{R_3^{\mathrm{A}}}{H_{\perp} R_2^{\mathrm{A}}} = \left(\frac{ea^2\nu}{2\kappa T}\right) F(T) e^{-(E_3-E_2-t)/\kappa T} \tag{13.31}$$

and

$$\mu_{\mathrm{t}}^{\mathrm{A}} = -\frac{3}{2} \frac{ea^2\nu}{\kappa T} e^{-(E_2-t)/\kappa T}, \tag{13.32}$$

where it is recalled that $A_{\Delta} = \sqrt{3}a^2/4$ for the inter-site separation a. Finally, these results are combined to yield the Hall coefficient for adiabatic semiclassical polaron hopping in the triangular lattice:

$$R_{H,\mathrm{t}}^{\mathrm{A}} = -\frac{\mu_{H,\mathrm{t}}}{\sigma} = \frac{1}{ne}\left(\frac{\mu_{H,\mathrm{t}}^{\mathrm{A}}}{\mu_{\mathrm{t}}^{\mathrm{A}}}\right) = \frac{-1}{ne}\left[\frac{F(T)}{3}\right] e^{(2E_2-E_3)/\kappa T}. \tag{13.33}$$

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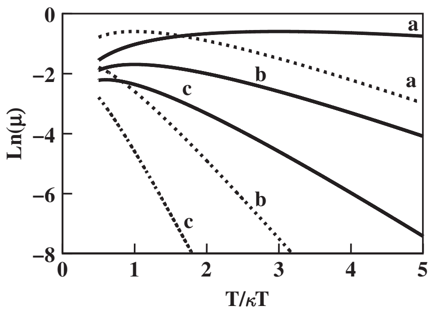

Fig. 13.10 The magnitudes of the triangular lattice's Hall mobilities (solid curves) and standard mobilities (dotted curves) are plotted in units of ea^2v/t against $t/\kappa T$. The three pairs of curves labeled as a, b and c denote progressively increasing values of the ratio of the E_2/t.

For the molecular-crystal model simple relationships exist among the deformation-related contributions to the characteristic energies of the Hall mobility, standard mobility and Hall constant: $E_3 - E_2 = E_2/3$ and $2E_2 - E_3 = 2E_2/3$.

The Hall-mobility activation energy of Eq. (13.31) is much smaller than the standard-mobility activation energy of Eq. (13.32). As a result the Hall mobility manifests a much weaker rise with increasing temperature than does the standard mobility. As illustrated in Fig. 13.10, the Hall mobility may even fall with increasing temperature. Except when the temperature is so high that the thermal energy overwhelms effects of the activation energies, the Hall mobility is larger than the conventional mobility.

The adiabatic approach applies when electronic transfer energies are large enough to preclude transitions between different adiabatic energy surfaces. By contrast, the non-adiabatic regime pertains when transitions between adiabatic surfaces dominate. As described in Section 11.6, the Landau–Zener analysis of the semiclassical transition between these two regimes indicates that it occurs when the transfer energy's magnitude becomes comparable to $t_m \sim (\hbar\omega)^{1/2}(E_2\kappa T)^{1/4}$. As expected, the standard mobilities for semiclassical polaron hopping calculated in the adiabatic and non-adiabatic limits, Eqs. (13.32) and (13.10), become comparable to one another when $t \sim t_m$. However, the ratio of the polaron-hopping Hall mobilities calculated in the semiclassical adiabatic and non-adiabatic limits, Eqs. (13.31) and (13.11), becomes $(t_m/\kappa T)\exp(t_m/\kappa T)F(T)$ when $t \sim t_m$. This ratio becomes much less than unity in the high-temperature limit where atomic motion becomes classical $t_m/\kappa T \ll 1$ (Emin and Holstein, 1969). Thus, the adiabatic Hall mobility is much less than the non-adiabatic Hall mobility in the transition regime $t \sim t_m$. This finding suggests that the Hall mobility does not monotonically rise as the transfer energy increases upon passing from the non-adiabatic to the adiabatic limits. Such non-monotonic behavior implies a complicated transition between the semiclassical Hall mobility's non-adiabatic and adiabatic limits.

The transition from non-adiabatic to adiabatic regimes is more complex for the semiclassical Hall mobility than for the standard mobility. In both regimes the standard mobility arises from trajectories which pass through a two-site coincidence. By contrast, different types of trajectory dominate the triangular lattice's Hall mobility in non-adiabatic and adiabatic regimes. In particular, the non-adiabatic Hall effect is dominated by trajectories which pass near the triple coincidence point (Friedman and Holstein, 1959). However the adiabatic Hall effect is dominated by trajectories which ride along the crest which defines a coincidence between a hop's two possible final sites (Emin and Holstein, 1969). Thus the transition between the semiclassical Hall mobility's non-adiabatic and adiabatic regimes accompanies a change of the weighting of these two different types of trajectory.

Finally, the three adiabatic energy surfaces for the three-site configuration, obtained from the eigenvalues of Eq. (13.17), are not equivalent with respect to inversion. This asymmetry is greatest near the origin. At the origin the energy of the lowest adiabatic surface is $\varepsilon(0,0) = -2t$ while the energies of the two higher-lying surfaces both equal t. Thus the adiabatic energy surface for a single excess electron hopping among otherwise vacant sites is not identical to that for a single unoccupied state hopping among sites that are otherwise occupied by electrons. Therefore the Hall-effect dynamics of a solitary excess electron hopping in a triangular lattice is not equivalent to that of a solitary vacant state. In particular, the degeneracy of two energy surfaces at the origin fosters non-adiabatic transitions between them. In addition, electronic degeneracy complicates the determination of the fictitious magnetic field associated with the higher-lying adiabatic surfaces. The t-dependences of the Hall-mobility activation energy and of the energy characterizing the Hall coefficient are also altered: cf. Eqs. (13.28), (13.31) and (13.33). Finally, the Hall-effect sign, addressed in Section 13.5, is not altered upon changing the carrier hopping in a triangular lattice from being an excess electron to being a vacant state (Holstein, 1973).

13.5 Hall-effect signs for hopping conduction

The Hall-effect sign indicates the nature of the charge carriers in a conventional wide-band semiconductor. Electrons that occupy states of positive effective mass near conduction-band minima produce a Hall coefficient of negative sign. By contrast, "holes", carriers associated with vacant electronic states having negative effective masses near valence-band maxima, produce a Hall coefficient of positive sign.

These simple relationships no longer apply when the widths of carriers' energy bands are less than the thermal energy κT. In particular, a different rule applies to

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carriers that move by hopping conduction. The sign of the Hall coefficient then depends on (1) the charge of the mobile specie, (2) the filling of the band of states amongst which carriers move, and (3) the signs of the electronic-transfer energies associated with the carriers' Hall effect.

This chapter's prior discussion of the Hall effect for polaron hopping explicitly considered a single excess electron hopping among sites that are empty. The complementary problem addresses the Hall effect for electronic hopping when all but a single site is occupied (Holstein, 1973). Passing from the empty-band limit to the full-band limit changes the sign of the Hall coefficient by the factor $(-1)^{n+1}$, where n denotes number of electronic transfers required to complete the closed loop associated with the Hall-effect interference process. With nearest-neighbor transfers $n = 3$ for the triangular lattice and $n = 4$ for the square lattice. Thus, the Hall coefficient for the triangular lattice does not change sign upon transforming from the hopping of a single excess electron to that of a single electronic vacancy: $(-1)^{n+1} = 1$ for $n = 3$. By contrast, the Hall coefficient for the square lattice changes sign upon transforming from the hopping of a single excess electron to that of a single electronic vacancy: $(-1)^{n+1} = -1$ for $n = 4$. This simple rule establishes the relative Hall-coefficient sign upon transforming from the hopping of an excess electron to that of an electronic vacancy.

Combining this rule with generalizations of the Hall coefficients obtained for excess electrons, described in the preceding sections of this chapter, yields a formula for the absolute sign of the Hall coefficient (Emin, 1977a):

$$\operatorname{sgn}(R_H) = \operatorname{sgn}\left[(-q)(\delta)^{n+1}\prod_{i=1}^{n} t_{i,i+1}\right]. \tag{13.34}$$

Here q denotes the sign of the bona fide carriers (usually electrons, $q = -e$). The filling of the band of states between which carriers hop is characterized by δ: $\delta = -1$ denotes a nearly empty band within which carriers move while $\delta = 1$ denotes a nearly filled band within which vacancies move. The final factor results from hopping conduction's Hall effect depending on the product of electronic transfer energies $t_{g,g+1}$ around a closed loop of n elements. Evaluation of Eq. (13.34) for simple models suggests explanations of why electronic hopping conduction often yields Hall coefficients with unconventional signs.

Consider electrons hopping ($q = -e$) between the nearly empty ($\delta = -1$) anti-bonding orbitals upon which conduction bands of common covalent semiconductors are based. A three-member closed loop of anti-bonding orbitals is schematically illustrated in panel a of Fig. 13.11. Each anti-bonding orbital is antisymmetric about the centroid of the line connecting the corresponding pair of neighboring atoms. The transfer integral that links adjacent anti-bonding orbitals garners its principal

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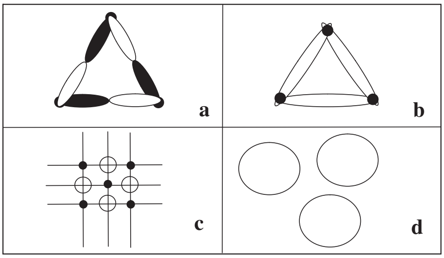

Fig. 13.11 Panel *a* depicts three atoms linked by three anti-bonding orbitals. The antisymmetric character of each of these orbitals is indicated by one of its two lobes being represented in black and the other lobe shown in white. The Hall coefficient for transport via such odd-membered loops is positive, independent of band filling. Panel *b* depicts three atoms linked by three bonding orbitals. The Hall coefficient for transport via such odd-membered loops is negative, independent of band filling. Panel c illustrates the structure of MnO. The small solid dots indicate the Mn cations and the large white circles denote the oxygen anions. The solid lines indicate that transfer between the cations is presumably facilitated by the intervening oxygen anion. The Hall coefficient is negative for nearly empty bands and positive for nearly filled bands. Panel *d* illustrates configurations of three mutual neighbors which usually dominate the Hall coefficient of randomly placed localized electronic states.

contribution from the region surrounding their mutually shared atom where these anti-bonding orbitals most strongly overlap. In this region the energy entering into the transfer integral is negative. In this circumstance the product of transfer energies for the closed loop is positive: e.g. $[(-1)\times(-1)]^n > 0$. Upon incorporating these results into Eq. (13.34) the sign of the Hall coefficient is found to be just the sign of δ^{n+1} with $\delta = -1$. Therefore nearest-neighbor hopping of excess electrons between anti-bonding orbitals arranged in odd-member loops (n is odd) generates a positive Hall coefficient.

Consider electrons hopping ($q = -e$) between the nearly filled ($\delta = 1$) bonding orbitals upon which highest-lying valence bands of common covalent semiconductors are based. A three-member closed loop of bonding orbitals is schematically illustrated in panel *b* of Fig. 13.11. With (1) the bonding orbitals being symmetric about the bond's mid-point, and (2) the energy entering into each transfer integral also being negative, the product of transfer energies is $(-1)^n$. Upon incorporating these results into Eq. (13.34) the sign of the Hall coefficient is also found to be just the sign of $(-1)^n$. Therefore nearest-neighbor hopping of an electronic vacancy between bonding orbitals arranged in odd-member loops (n is odd) generates a negative Hall coefficient.

Hall-effect sign anomalies have been reported in amorphous covalent semiconductors for which there is evidence of hopping conduction. Positive Hall

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coefficients are reported when both amorphous silicon and amorphous arsenic have been doped to introduce electrons into their conduction states (LeComber *et al.*, 1977; Mytilineou and Davis, 1977). In addition, negative Hall coefficients are reported when amorphous germanium and amorphous silicon are doped to remove electrons from their valence states (Clark, 1967; Seager *et al.*, 1974; LeComber *et al.*, 1977). Unlike their crystalline versions, the structures of these amorphous covalent semiconductors are thought to contain significant numbers of odd-member rings (Thorpe *et al.*, 1973; Joannopolous and Cohen, 1973; Robertson, 1975; Paul and Connell, 1976).

Hopping also occurs between localized electronic states that are not involved in covalent bonding. These sites can be those of self-trapped carriers in crystals. Examples include individual cation sites in some transition-metal compounds (e.g. Mn sites in MnO) and sites of V_K-centers (Cl_2^- units) for self-trapped holes in KCl. A portion of the MnO lattice is schematically depicted in panel *c* of Fig. 13.11. Hopping also takes place among non-bonding states in non-crystalline solids. Examples of these include lone-pair states associated with the uppermost valence bands in a number of glasses. In addition, electronic hops proceed between defect states and among impurity states including donor states in lightly doped conventional semiconductors. A non-crystalline arrangement of the orbitals of three mutual neighbors is schematically shown in panel *d* of Fig. 13.11.

Equation (13.34) gives the sign of the Hall coefficient in each of these instances of electronic hopping ($q = -e$) as just the sign of $\delta^{n+1}(-1)^n$, where (as with bonding orbitals) the final factor is the sign of the product of the n transfer energies in the pertinent loop. Thus the Hall coefficient is positive for hopping between nearly filled (δ = 1) nearest-neighbor sites of a cubic crystal (n = 4). For example, the hopping of electronic vacancies introduced by very sparse substitution of Li for Mn in cubic MnO produces a positive Hall coefficient (Crevecoeur and de Wit, 1970). By contrast, the Hall coefficient is negative for hopping among randomly placed sites whose predominant loops link three mutually adjacent neighbors (n = 3). Indeed, hopping in chalcogenide glasses often produces negative Hall coefficients even when the Seebeck coefficients are positive (Emin *et al.*, 1972; Seager *et al.*, 1973, Grant *et al.*, 1974; Nagels *et al.*, 1974; Klaffke and Wood, 1976).

Finally, a situation is noted in which the bona fide carriers are holes ($q = e$). At very low temperatures holes in doped conventional semiconductors are bound in *s*-like shallow acceptor states. At low enough acceptor densities these holes can hop between acceptors. The sign of the Hall coefficient associated with this hopping conduction is then $-\delta^{n+1}(-1)^n$. With three-member loops dominating the Hall effect at low acceptor concentrations, the Hall coefficient is always positive. In particular,

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the Hall coefficient remains positive even though the Seebeck coefficient will change sign with compensation (e.g. changing δ from -1 to 1). An analogous situation occurs for inter-donor hopping ($q = -e$) in conventional semiconductors. The Hall coefficient then remains negative while the Seebeck coefficient changes sign with compensation (Geballe, 1959).